

Convergent Discovery of Critical Phenomena Mathematics Across Disciplines*

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Abstract

Techniques for detecting critical phenomena—phase transitions where correlation length diverges and small perturbations have large effects—have been developed across multiple fields over nine decades. We survey between six and twelve disciplines (depending on classification criteria) where researchers derived functionally corresponding measures of correlation scaling, with little documented awareness of each other’s work. The physicist’s correlation length ξ , the cardiologist’s DFA scaling exponent α , the financial analyst’s Hurst exponent H , and the machine learning engineer’s spectral radius χ all detect critical signatures under different notation.

We classify each surveyed domain as independent derivation, domain transfer, or empirical precursor, and present citation network evidence that cross-domain citations remained significantly below random-mixing expectations (under a simple null model of domain mixing) during the formative period (1987–2010). The framework that motivated this investigation—derived from distributed systems engineering and presented in the Supplementary Material—is excluded from the survey count given the authors’ dual role.

Building on prior syntheses—notably Sornette’s 2004 textbook—this paper contributes a taxonomy of discovery types and quantitative documentation of the convergence pattern.

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Plain Language Summary

Plain language title: Scientists in different fields independently invented the same failure-prediction math

When a complex system—a heart, a power grid, a financial market, a climate system—approaches a catastrophic transition, its behavior changes in predictable ways. Small disturbances take longer to die out. Distant parts of the system begin to move in sync. Fluctuations grow. Scientists have developed mathematical tools to detect these warning signs and predict when a system is nearing a tipping point.

What is remarkable is that researchers in at least six different fields invented essentially the same mathematics for this purpose, over nine decades, mostly without knowing about each other’s work. A physicist measuring “correlation length,” a cardiologist computing a “DFA scaling exponent,” a financial analyst calculating a “Hurst exponent,” and a machine learning engineer tuning a “spectral radius” are all answering the same question: how close is this system to a critical transition? Their tools detect the same signatures—growing correlations, slowing recovery, increasing sensitivity—using different notation developed independently within each discipline.

This paper surveys between six and twelve fields (depending on how strictly one counts) where this convergent discovery occurred. We classify each case as independent invention, adaptation from another field, or an early empirical observation that predated the formal mathematics. We also analyze citation networks—who cited whom—and find that cross-field awareness was remarkably low during the key period from 1987 to 2010, even though the techniques were functionally equivalent.

The convergence has practical implications. Insights from one domain can transfer to another: a power grid engineer can learn from how cardiologists detect cardiac risk, and climate scientists’ tipping-point models share mathematical structure with financial crash prediction. Documenting this pattern is a step toward breaking down the disciplinary barriers that kept these communities working in parallel for decades.

Introduction

The Convergence Pattern

Critical phenomena exhibit universality: systems with vastly different microscopic dynamics display identical scaling behavior near continuous phase transitions. The two-point correlation function

$$C(r) \sim r^{-(d-2+\eta)} e^{-r/\xi} \quad (1)$$

characterizes fluctuations at separation r , with correlation length ξ diverging at the critical point as $\xi \sim |T - T_c|^{-\nu}$, where T is the control parameter (e.g., temperature) and T_c is its critical value at which the phase transition occurs. The critical exponents ν , η depend only on dimensionality and symmetry class—not microscopic details. This universality, established through renormalization group theory, is among the deepest results in statistical mechanics.

We document a recurring pattern of convergence in the discovery process itself: researchers across multiple disciplines derived functionally corresponding measures of correlation scaling—often without awareness of each other’s work. A physicist measuring ξ , a cardiologist computing detrended fluctuation analysis (DFA) scaling exponent α , a quant calculating Hurst exponent H , and a machine learning engineer tuning spectral radius χ are performing functionally equivalent calculations under different notation.

The physics of universality—that systems in the same symmetry class share critical exponents—is well established. Our contribution is different: we document the *discovery process* by which practitioners outside physics independently arrived at functionally corresponding diagnostics, classify each case, and present quantitative citation evidence bearing on whether this reflects knowledge transfer or independent derivation. The sociology of science has long recognized simultaneous independent discovery as a recurring phenomenon ([Merton 1961](#)); we provide a specific, quantitatively documented case. Section presents the citation network evidence.

Historical Context

Statistical physics established the mathematical framework between 1944 and 1971: Onsager’s exact Ising solution ([Onsager 1944](#)), Kadanoff’s block spin renormalization ([Kadanoff 1966](#)), and

Wilson’s renormalization group theory ([Wilson 1971](#)) (Nobel Prize 1982), comprehensively reviewed by Fisher ([Fisher 1974](#)). By the early 1970s, this was celebrated, publicly available science.

Why did it take 15–50 years for equivalent techniques to appear elsewhere? The most parsimonious explanation is natural convergence: as complexity science, biomedicine, finance, and machine learning matured during the 1980s–2000s, researchers confronting correlation problems derived similar mathematics from first principles. Disciplinary barriers compound this—a physicist’s “correlation length” and a financial analyst’s “Hurst exponent” look like different things even when they diagnose the same critical regime.

The pattern predates formal criticality theory. In the mid-twentieth century, three scientists in unrelated fields independently discovered power-law scaling: [Gutenberg and Richter \(1944\)](#) characterized earthquake frequency-magnitude distributions, [Richardson \(1960\)](#) found power-law scaling in war casualties (published posthumously 1960), and [Zipf \(1949\)](#) documented word-frequency power laws in natural language (1935–1949). None cited the others. The Hurst exponent—now foundational across finance, biomedicine, and physics—originated when hydrologist Harold [Hurst \(1951\)](#) discovered long-range dependence in Nile River discharge data; Mandelbrot formalized it mathematically seventeen years later, and it was independently rediscovered as a diagnostic tool in physiology (Peng’s DFA ([Peng et al. 1994](#))) and finance (Peters ([Peters 1994](#))).

The nonlinear dynamics community recognized universality across chaotic systems by the mid-1980s ([Cvitanović 1984](#)); that applied communities continued independent derivation despite this recognition strengthens the convergence thesis. Similarly, Sethna et al.’s “crackling noise” framework ([Sethna et al. 2001](#)) unified Barkhausen noise in magnets (discovered 1919), earthquake avalanches, and fracture mechanics under a single universality class—demonstrating that identical scaling functions emerge across systems with completely different microscopic physics.

A notable prior synthesis is Sornette’s 2004 textbook ([Sornette 2004](#)), which explicitly unified critical phenomena across natural sciences—applying statistical mechanics concepts to biology, economics, and geophysics. [Scheffer et al.](#)’s influential review ([Scheffer et al. 2009](#)) further demonstrated cross-domain applicability of early warning signals. That domain-specific literatures continued developing largely in parallel despite these works suggests that even high-quality synthesis faces adoption barriers across disciplinary boundaries. Our contribution is not the first cross-domain synthesis but rather quantitative documentation of the convergence pattern: tracing

when and how awareness spread, and classifying each surveyed instance by type.

Our investigation was prompted by a specific case of apparent independent derivation (see Supplementary Material): a framework developed in distributed systems engineering whose critical-point behavior resembled established physics. We document the broader convergence pattern the resulting literature search uncovered.

Definitions and Scope

To structure the evidence in Section , we adopt five classification categories:

Independent derivation: Framework developed from first principles without demonstrable awareness of equivalent prior work.

Domain transfer: Existing technique applied to a new domain with traceable lineage.

Intra-field extension: New phenomena identified using existing tools within the same discipline.

Empirical precursor: Empirical observation of critical-like behavior predating the formal framework.

Foundational cross-domain contribution: General mathematical framework (scaling, self-similarity, power laws) influencing multiple discovery lineages.

We use “qualified” to indicate cases where the mathematical method itself is novel but the researcher’s training or institutional context includes significant exposure to the relevant physics, creating a partial intellectual lineage that falls short of direct methodological borrowing.

We use *functional correspondence* to describe the relationship among the parameters documented here: two parameters exhibit functional correspondence if they identify the same system states as critical and non-critical. These measures detect the same critical signatures—diverging correlation, critical slowing down, power-law scaling—but are not in all cases formally interconvertible. For stationary long-memory processes, exact mathematical equivalence holds between certain pairs (Section demonstrates one such chain). For others, the correspondence is phenomenological rather than algebraic.

The scope of this paper is a historical survey of convergent discovery, supported by citation network analysis. We do not claim mathematical proof of universal equivalence; we document a pattern and present evidence bearing on whether it reflects independent derivation or knowledge transfer.

Table 1: Classification of discoveries surveyed in Section

Domain	Classification	Notes
Statistical Physics	Foundational framework	Established the mathematics
Seismology (G-R 1944)	Empirical precursor	Predates SOC by 43 years
SOC (Bak 1987)	Intra-physics extension	Physicist applying stat-mech
Edge of Chaos (Kauffman)	Qualified independent	Biologist; SFI exposure
Biomedical (DFA)	Qualified independent	Stanley’s physics background
Finance (Mandelbrot 1963)	Foundational cross-domain	Fractal geometry; scaling laws
Finance (Peters 1994)	Domain transfer	Applied existing technique
Machine Learning	Qualified independent	ESNs; edge-of-chaos circulating
Neuroscience	Qualified independent	Beggs & Plenz; cited Bak
Power Grids	Independent derivation	Cascade failure analysis
Traffic Flow	Empirical precursor + qualified	Greenshields / Kerner
Climate / Thermohaline	Independent derivation	Bifurcation analysis of AMOC
Linguistics (Zipf)	Empirical precursor	Power-law word frequency
Urban Scaling	Domain transfer	From metabolic scaling theory
Sornette (2004)	Prior cross-domain synthesis	Textbook treatment

Between six and twelve of the domains surveyed represent convergent discovery, depending on how strictly one defines “independent derivation.” Six domains—Kauffman’s edge of chaos, biomedical signal analysis, neuroscience, machine learning, power grid analysis, and traffic flow (Kerner)—present the clearest cases for independent development, though with qualifications noted for each. Including domain transfers (finance, urban scaling), empirical precursors (seismology, linguistics), and climate/thermohaline circulation expands the count to twelve (Table 1). Bak’s SOC and Sornette’s prior synthesis are classified separately. The authors’ own framework (Supplementary Material), which motivated this survey, is presented separately and excluded from this count.

Cross-Domain Discoveries

Statistical Physics (1944–1971)

Classification: foundational framework.

Onsager’s exact solution to the 2D Ising model (Onsager 1944) identified the critical temperature $T_c = 2.269$ and showed that correlation length diverges as $\xi \sim |T - T_c|^{-\nu}$ near criticality.

Complexity Science (1987–1993)

Classification: Bak et al. (Bak et al. 1987)—intra-physics extension. Kauffman (Kauffman 1993)—qualified independent derivation (biologist with Santa Fe Institute exposure to physics; different mathematical foundations).

Bak, Tang, and Wiesenfeld (1987) introduced self-organized criticality (SOC) (Bak et al. 1987), showing that complex systems spontaneously evolve toward critical states. Avalanche sizes follow power laws $P(s) \sim s^{-\tau}$ —the same divergence of correlation structure that characterizes continuous phase transitions. Kauffman’s “Origins of Order” (Kauffman 1993) identified the edge of chaos in Boolean networks: ordered ($K < 2$), critical ($K \approx 2$), chaotic ($K > 2$). Maximal computational capability occurs at $K = 2$, where correlation length diverges— K serves as a tuning parameter analogous to temperature.

Biomedical/HRV Analysis (1994)

Classification: qualified independent derivation. Peng and collaborators derived DFA from first principles for DNA sequence analysis, but the group was led by H. Eugene Stanley, a statistical physicist. The methodology is original; the intellectual lineage to physics is not fully severed.

Peng et al. introduced Detrended Fluctuation Analysis (DFA) for analyzing DNA sequences (Peng et al. 1994). The scaling exponent α reveals correlation structure:

$$F(n) \sim n^\alpha \tag{2}$$

$$\alpha \approx 0.5 \implies \text{uncorrelated (white noise)} \tag{3}$$

$$\alpha \approx 1.0 \implies \text{critical (long-range correlation)} \tag{4}$$

$$\alpha > 1.0 \implies \text{over-correlated} \tag{5}$$

Financial Markets (1990s)

Classification: Mandelbrot (1963)—foundational cross-domain contribution. Peters (1994)—domain transfer. Mandelbrot’s 1963 analysis of speculative prices applied Hurst’s R/S technique to financial data, but his broader contribution was fractal geometry as a framework for self-similar phenomena across scales and domains. His promotion of scaling laws and power-law distributions shaped the

intellectual environment from which several later discoveries in this survey emerged—particularly through the statistical physics community (Stanley, Mantegna) that produced both econophysics and biomedical DFA. Peters extended this application using the Hurst exponent.

Peters (1994) applied fractal analysis to financial markets using the Hurst exponent. Originally developed by Hurst (1951) for Nile River hydrology, H characterizes long-term memory in time series. Mandelbrot (1963) first applied Hurst’s R/S analysis to financial time series, establishing the connection between hydrology and market dynamics decades before Peters.

The Hurst exponent comes from rescaled range (R/S) analysis:

$$\frac{R(n)}{S(n)} \sim n^H \quad (6)$$

where R is the range of cumulative deviations and S is standard deviation over window size n :

- $H = 0.5 \implies$ random walk (uncorrelated, efficient market)
- $H > 0.5 \implies$ trending/persistent (positive correlation)
- $H < 0.5 \implies$ mean-reverting (negative correlation)
- $H \approx 1.0 \implies$ critical regime (long-range correlation)

Real markets typically show $H \approx 0.7$ – 0.8 . During market stress, H approaches 1.0—indicating diverging correlation length and the onset of collective behavior across otherwise independent market participants. Mantegna and Stanley (1995) subsequently linked econophysics to statistical mechanics through power-law return distributions.

Machine Learning (2001–2017)

Classification: qualified independent derivation. Jaeger (2001) does not cite SOC or Kauffman, but the edge-of-chaos concept was circulating in neural computation by the late 1990s.

Jaeger’s 2001 Echo State Networks (ESNs) revealed that recurrent neural networks operate optimally at the “edge of chaos” (Jaeger 2001). The key parameter is spectral radius χ (largest eigenvalue) of the weight matrix:

$$\chi = \max |\lambda_i(\mathbf{W})| \quad (7)$$

ESN dynamics show three regimes:

- $\chi < 1 \implies$ contracting (ordered, stable, limited memory)
- $\chi \approx 1 \implies$ critical (edge of chaos, optimal computation)
- $\chi > 1 \implies$ expanding (chaotic, unstable)

At $\chi \approx 1$, networks achieve maximum computational capability through critical slowing down—the same phenomenon causing $\tau \rightarrow \infty$ at phase transitions. Perturbations persist longer, creating temporal memory for sequence processing. Jaeger’s original ESN technical report (Jaeger 2001) does not cite Bak, Kauffman, Langton, or the SOC/edge-of-chaos literature, focusing instead on recurrent network training within the neural computation community.

Saxe et al. (2013) extended this to deep feedforward networks, showing that optimal weight initialization requires operating near the order-chaos boundary. Information propagates through layers most effectively at criticality; away from it, signals either attenuate or diverge.

Neuroscience (2003)

Classification: qualified independent derivation. Beggs and Plenz discovered power-law avalanche statistics empirically in cortical tissue; they cited Bak et al. but their finding was driven by neural recording data, not by applying SOC theory.

Beggs and Plenz (2003) recorded local field potentials in rat cortical slices and found that spontaneous activity propagates as neuronal avalanches with size distribution $P(s) \sim s^{-3/2}$ —matching the mean-field critical branching prediction. At criticality, neural networks achieve maximal dynamic range, optimal information transmission, and the largest repertoire of activity patterns.

The “criticality hypothesis”—that the brain operates near a critical point for computational advantage—has since become a major research program. DFA applied to EEG and MEG data connects directly to the biomedical scaling analysis of Section . As with DFA in cardiology, the approach converged on criticality mathematics from domain-specific empirical observation rather than from deliberate application of statistical mechanics.

Power Grid Analysis (2000s–2010s)

Classification: independent derivation.

Power grid engineers arrived at critical phenomena mathematics through cascade failure anal-

ysis. [Crucitti et al. \(2004\)](#) showed that blackout cascades exhibit self-organized criticality, with blackout sizes following power laws $P(s) \sim s^{-\alpha}$. [Dobson et al. \(2007\)](#) went further, demonstrating that grids naturally evolve toward critical operating points as demand increases and margins shrink.

The key observable is generator coherence. As load approaches the critical threshold, distant generators become increasingly synchronized—spatial correlation ξ grows. Engineers developed early warning indicators from this: system recovery time τ increases as criticality approaches, a direct application of critical slowing down.

Traffic Flow (1935, 2010s)

*Classification: Greenshields (1935) is an empirical precursor—he observed phase-transition-like behavior decades before the formal framework existed.*¹

Greenshields’ 1935 study ([Greenshields 1935](#)) identified a fundamental relationship between traffic density ρ and flow rate, predating the formal mathematical framework by decades. Traffic shows distinct phases—free flow at low density, congestion at high density—with the transition at critical density ρ_c . Greenshields recognized this empirically; the explicit criticality framing came later through Kerner and others.

Kerner’s three-phase model (2004) ([Kerner 2004](#)) formalized this as a genuine phase transition: free flow ($\rho < \rho_c$), synchronized flow at criticality ($\rho \approx \rho_c$), and wide moving jams ($\rho > \rho_c$). Near ρ_c , jam sizes follow power laws and correlation length diverges—disturbances propagate arbitrarily far upstream. The three-regime structure recurs here without methodological borrowing from the fields that formalized it first.

Thermohaline Circulation / Climate Tipping Points (1961–2021)

Classification: independent derivation within climate science; connected to physics through shared bifurcation theory but developed for domain-specific physical systems without awareness of equivalent biomedical or financial applications.

[Stommel \(1961\)](#) introduced a two-box model of thermohaline circulation exhibiting bistability—the Atlantic Meridional Overturning Circulation (AMOC) can occupy either a strong or collapsed

¹Kerner (2004) provided the explicit criticality formalization. Kerner’s classification is qualified independent: his three-phase model draws on statistical mechanics concepts, though developed primarily within the traffic engineering community.

equilibrium, with hysteresis preventing smooth recovery. This was among the earliest applications of bifurcation analysis to a geophysical system.

[Schmidt and Mysak \(1996\)](#) demonstrated that zonally averaged thermohaline models exhibit a pitchfork bifurcation—the symmetric two-cell circulation loses stability as diffusivity increases, giving way to asymmetric one-cell patterns. [Wang and Mysak \(2006\)](#) showed that Dansgaard-Oeschger oscillations in paleoclimate records correspond to stochastic switching near a Hopf bifurcation in the meridional overturning circulation, with irregular periods characteristic of critical slowing down—the same signature that DFA detects in cardiac time series.

[Rahmstorf et al. \(2005\)](#), in an intercomparison of eleven independent climate models, found that all exhibited the same hysteresis loop for AMOC response to freshwater forcing—convergent discovery *within* a single domain, paralleling the cross-domain convergence this paper documents. [Lenton et al. \(2008\)](#) synthesized these results into a tipping elements framework for the climate system, citing both Rahmstorf and the broader bifurcation literature.

The chain from climate science to the methods surveyed here closes through [Dakos et al. \(2012\)](#), who provided a systematic toolbox for detecting early warning signals across ecological and climate systems, and [Bury et al. \(2021\)](#), who trained deep learning classifiers on normal form bifurcation data—saddle-node, Hopf, transcritical—and demonstrated that these classifiers generalize across ecology, climate, and cardiology without domain-specific retraining. [Bury et al. \(2023\)](#) validated the approach on experimental chick-heart aggregate data—a cardiac system originally studied by Glass and collaborators ([Mackey and Glass 1977](#))—using training data from the same bifurcation normal forms that govern AMOC tipping points. Bury’s lineage traces through [Scheffer et al.’s \(2009\)](#) bridging review to [Wissel \(1984\)](#), who documented characteristic return time near thresholds in ecology—a pre-Scheffer articulation of critical slowing down—and to [Gardiner’s \(1983\)](#) stochastic processes handbook, which formalized the mathematical tools both ecologists and physicists would independently adopt. The mathematical structure connecting these domains was always there; practitioners in each field developed their diagnostics independently.

A concrete illustration: Mysak and Glass are both emeritus professors at McGill University, both Fellows of the Royal Society of Canada, and both played in the same faculty orchestra. Mysak has noted that he “sometimes meet[s] Leon Glass on the bus and then I can talk about our ‘related’ work” (personal communication, 2026). They did not know their research used the same

bifurcation mathematics until an outside investigator connected them. The convergence pattern this paper documents is not an abstraction—it persists even at the scale of a single university bus route.

Seismology (1944)

Classification: empirical precursor. Gutenberg-Richter predates SOC by 43 years; Bak later claimed earthquakes as SOC.

Gutenberg and Richter (1944) established that earthquake frequency decreases as a power law of magnitude: $\log_{10} N = a - bM$, with $b \approx 1$. This is arguably the cleanest natural power law ever measured. Bak and Tang (1989) later interpreted seismicity as self-organized criticality, but the empirical scaling was discovered four decades earlier by seismologists with no connection to statistical physics.

Linguistics (1935–2003)

Classification: empirical precursor (Zipf); qualified independent (Ferrer i Cancho).

Zipf (1949) documented power-law word-frequency distributions in natural language. Ferrer i Cancho and Solé (2003) argued that Zipf’s law emerges at a phase transition between communicatively useless randomness and rigid one-to-one coding—natural language, on this account, operates at a critical point optimizing the trade-off between speaker effort and listener ambiguity. The mathematical apparatus (information-theoretic optimization yielding power-law statistics at a phase boundary) was developed without reference to physics criticality.

Urban Scaling (2007)

Classification: domain transfer from metabolic scaling theory, with novel empirical content.

Bettencourt et al. (2007) discovered that cities obey universal power-law scaling: infrastructure scales sublinearly with population ($\sim N^{0.85}$, economies of scale) while innovation and wealth scale superlinearly ($\sim N^{1.15}$, increasing returns). These exponents are remarkably consistent across countries and centuries. The mathematical framework draws on West’s metabolic scaling theory

(itself rooted in physics), but the empirical regularities—and the sub/superlinear distinction—were discovered in urban data.

Evidence for Independent Discovery

Cross-Citation Evidence

If techniques spread through knowledge transfer, we would expect cross-domain citations. Instead, citation analysis (via the Semantic Scholar API) reveals citation patterns significantly more domain-clustered than random mixing would predict, in every period analyzed. All domains cite foundational physics (Onsager, Wilson) in background sections, but not as primary methodological sources.

During the formative period (1987–2010), cross-domain citations are conspicuously absent. Peters (1994, finance) does not cite Peng et al. (1994, biomedical)—published the same year, using equivalent techniques. Jaeger (2001, ESNs) does not cite Kauffman (1993, Boolean networks)—same phenomenon, different notation. The power grid literature develops without citing self-organized criticality. Traffic researchers cite Greenshields (1935) but not SOC, DFA, or Hurst analysis.

The notable exception—Kauffman (1993) citing Bak et al. (1987)—still shows no awareness of simultaneous biomedical or financial applications. Sornette’s 2004 textbook (Sornette 2004) explicitly synthesized critical phenomena across natural sciences, yet domain-specific literatures continued developing largely in parallel afterward. That even high-quality synthesis failed to prevent continued parallel development strengthens the case that convergent derivation, not knowledge transfer, drove the pattern.

A clear contrast case is Moore and Mertens’ work on phase transitions in computational complexity (Moore and Mertens 2011). Unlike the independent derivations above, Moore—a physicist at the Santa Fe Institute—explicitly applied statistical mechanics to random k -SAT problems. This deliberate cross-pollination illustrates what knowledge transfer looks like, contrasting sharply with the absent citation trails of 1987–2010.

Cross-domain awareness emerges primarily after 2010: ML papers begin citing complexity science, econophysics bridges finance and physics. This delayed integration supports natural

convergence—each domain developed independently, with cross-fertilization coming after maturation.

Quantitative citation analysis confirms this pattern. We selected seed papers as the foundational or most-cited methodological contribution in each domain (e.g., Onsager for physics, Peng for DFA, Peters for finance)—choosing papers that defined a technique rather than papers likely to attract cross-domain attention. Analyzing forward citations of these 15 seed papers across six Semantic Scholar-resolvable domains (28,591 unique citing papers), we find cross-domain citation rates consistently below the random-mixing null model in every period ($p < 0.0001$; Table 2). Cross-domain rates rise modestly from 64% (1996–2005) to 69% (2006–2015) before declining to 56% (2016–2026); the recent decline likely reflects the high rate of unclassified papers in Semantic Scholar’s recent records (19% unknown in 2016–2026 vs. 4% in 1996–2005). The consistently siloed pattern—observed rates 15–20 percentage points below random-mixing expectations—supports independent development rather than cross-pollination.

Table 2: Cross-domain citation rates for 15 seed papers across 6 domains

Period	N	Observed	Expected (null)	χ^2
1996–2005	2,281	63.8%	80.5%	402.8***
2006–2015	9,969	68.8%	84.1%	1,754.8***
2016–2026	14,110	55.5%	75.5%	3,049.3***

*** $p < 0.0001$. Null model: expected cross-domain rate = $1 - \sum p_i^2$ (Herfindahl index) where p_i is domain share. Data from Semantic Scholar API (28,591 unique citing papers). Two papers truncated at 62–66% of total citations due to API rate limits. The 1987–1995 period ($N = 34$) is omitted due to insufficient sample size. Papers with no Semantic Scholar field classification (13.6% overall, 19% in 2016–2026) are excluded; see sensitivity analysis in text.

This null model assumes unrestricted mixing across domains—a deliberate simplification. In practice, journal scope constraints, reviewer pool segmentation, and field-specific citation norms reduce expected cross-citation rates below the random-mixing baseline. A more restrictive null—for example, a stochastic block model conditioned on journal co-occurrence or a gravity model incorporating disciplinary distance—would likely still show significant clustering but with reduced effect sizes. We therefore interpret the chi-squared results as evidence for domain clustering consistent with the qualitative evidence (absent citations, independent notation, institutional separation), while acknowledging that the magnitude of departure from the HHI expectation is an upper bound on the “true” clustering effect.

To bound the effect of unclassified papers (13.6% overall, 19% in 2016–2026), we computed

worst- and best-case cross-domain rates by assigning all unknowns to same-domain or cross-domain respectively: the 2016–2026 rate ranges from 45.2% to 63.8%, compared to 55.5% observed, and all periods remain significantly below the null expectation. Merging Biology and Biomedical into a single domain—the most aggressive reasonable consolidation—reduces cross-domain rates modestly (e.g., 55.5% $\rightarrow$ 43.7% in 2016–2026) but preserves significant siloing in every period. The core finding is robust to both perturbations.

Notably, Sornette’s cross-domain synthesis textbook (Sornette 2004) was itself primarily cited within physics (cross-domain rate: 36%), suggesting that even explicit synthesis faces adoption barriers. Domain classifications use Semantic Scholar’s algorithmic `fieldsOfStudy` mapped to six broad categories. These classifications are imperfect—S2 assigns fields based on venue and content signals, not expert curation—but represent the best available large-scale classification. We validated spot-checks against manual inspection and found reasonable agreement for the broad categories used here. Additional seed papers from seismology, linguistics, urban science, and other sub-domains were excluded from the quantitative table due to missing field classifications but contribute narrative evidence in Section .

Notation Divergence

Each domain developed its own notation for the same underlying mathematics, with no terminological inheritance between them. Statistical physics measures correlation length ξ and critical exponents ν , η . In cardiology, DFA introduced scaling exponent α ; financial analysts adopted the Hurst exponent H from hydrology. Machine learning settled on spectral radius χ for recurrent network dynamics. Traffic engineers characterize transitions through critical density ρ_c .

None of these notation systems show terminological inheritance from each other—a physicist would not recognize α as diagnosing the same critical regime as ξ without careful analysis, nor would a financial analyst connect H to χ . This contrasts with deliberate transfer cases: Moore and Mertens (2011), applying phase transition theory to computational complexity, explicitly preserved the statistical mechanics language of “order parameters” and “critical thresholds.” Notation divergence is the expected signature of independent derivation; notation preservation signals knowledge transfer. The contrast with calculus is instructive: Leibniz’s and Newton’s distinct notations eventually consolidated into shared vocabulary across every field. No comparable consolidation has

occurred for criticality (cf. Bury, personal communication, 2026).²

Institutional Separation

The surveyed discoveries span different countries, institutions, and funding ecosystems. Bak developed SOC at Brookhaven National Laboratory while Peng’s group at Boston University created DFA for biomedical signals. Peters worked as an independent practitioner in finance; Jaeger introduced echo state networks at the University of Bremen. Crucitti’s cascade failure analysis emerged from the University of Catania, and Dobson’s power grid work from Iowa State. No common funding sources, advisory chains, or conference networks connected these researchers’ criticality work during the formative period.

The principal exception is the Boston University physics department under H. Eugene Stanley, whose group produced both DFA methodology (Peng et al. 1994) and early econophysics contributions (Mantegna and Stanley 1995)—hence our classification of DFA as “qualified independent.” Conference programs from the 1990s show no cross-domain presentations; co-authorships bridging these domains appear only after 2010.

Functional Correspondence

Core Mathematical Structure

All domains measure correlation decay rates:

$$C(r) \sim r^{-\alpha} \quad (\text{spatial}) \quad C(t) \sim t^{-\beta} \quad (\text{temporal}) \quad (8)$$

At criticality, these exponents approach values indicating long-range correlation.

Parameter Mapping

**Note on HRV:* Healthy cardiac dynamics operate *away from* criticality ($\alpha_1 \approx 0.75$); values of $\alpha_1 \approx 1.0$ indicate pathological loss of autonomic regulation (Peng et al. 1995). DFA thus serves

²Bury (personal communication, 2026) notes that practitioners moving between fields must learn each domain’s vocabulary independently.

Table 3: Functional correspondence of critical phenomena parameters across domains

Domain	Parameter	Common name in domain	Meaning	Critical Value
Physics	ξ	Correlation length	Spatial correlation decay	$\xi \rightarrow \infty$
Physics	τ	Relaxation time	Temporal correlation decay	$\tau \rightarrow \infty$
DFA	α	DFA exponent	Detrended fluctuation scaling	$\alpha \approx 1$
Finance	H	Hurst exponent	Long-range dependence	$H \approx 1$
ML	χ	Spectral radius	Recurrent weight stability	$\chi \approx 1$
HRV	α_1	Short-term DFA exponent	Autonomic regulation index	$\alpha_1 \approx 0.75^*$
Traffic	ρ_c	Critical density	Flow-density phase boundary	Phase transition
<i>Empirical scaling exponents (power-law distributions, not correlation-decay diagnostics):</i>				
Seismology	b	b -value (Gutenberg–Richter)	Frequency-magnitude scaling exponent	$b \approx 1$
Linguistics	γ	Zipf exponent	Word rank-frequency scaling	$\gamma \approx 1$
Urban science	β	City scaling exponent	Superlinear scaling with population	$\beta > 1$ (innovation)

the same diagnostic function while the clinically relevant value differs because healthy hearts avoid sustained criticality.

Notes: We use “functional correspondence” rather than “mathematical equivalence” because these parameters detect the same critical signatures without necessarily being interconvertible across all process types (see Section). These parameters fall into distinct mathematical categories: ξ and τ are diverging correlation measures; α and H are global scaling exponents; χ is a local linear stability condition; ρ_c is a threshold parameter. The last three entries (b , γ , β) are empirical scaling exponents from power-law distributions rather than correlation-decay diagnostics; they are included to show the breadth of critical-point mathematics across domains.

Rigorous Equivalence and the Unifying Principle

For at least one pair of parameters, the correspondence is not merely functional but mathematically exact. For fractional Gaussian noise (fGn) with long-range dependence parameter $H \in (0, 1)$, the DFA scaling exponent satisfies $\alpha = H$ (Kantelhardt et al. 2002). This identity extends to the spectral domain: for processes with power-spectral density $S(f) \sim f^{-\beta}$, both parameters relate through

$$\beta = 2H - 1 = 2\alpha - 1 \quad (9)$$

establishing mathematical equivalence—not merely functional correspondence—for stationary long-memory processes. At the critical point ($\beta = 1$), both H and α equal unity and the process exhibits $1/f$ noise, the spectral signature of criticality.

For non-stationary processes (fractional Brownian motion), the relationship shifts to $\alpha = H + 1$, but remains exact and well-defined. The existence of rigorous equivalence chains for specific process classes suggests that the broader functional correspondences documented in Table 3 reflect genuine mathematical structure rather than superficial analogy. We do not claim this equivalence generalizes beyond the fGn/fBm family; it serves as an existence proof that at least some cross-domain correspondences are exact, motivating further investigation of others.

All techniques diagnose *correlation decay rate*: slow decay ($\alpha \approx 1$, $H \approx 1$, $\chi \approx 1$, large ξ , τ) signals long-range correlations and the approach to criticality; fast decay ($\alpha \approx 0.5$, $H = 0.5$, $\chi < 1$) signals absence of criticality. Whether measuring spatial extent (ξ), temporal persistence (τ), scaling exponents (α , H), dynamical measures (χ , K), or thresholds (T_c , ρ_c)—all answer the same question: how far do correlations extend?

Systems near criticality exhibit optimal information processing, predictive early warning signals via critical slowing down, universal behavior independent of microscopic details, and extreme sensitivity to perturbation. The convergence across disciplines is not coincidental—correlation decay is the natural observable for any system with phase transitions, and the mathematics is dictated by the structure of the problem.

Convergence Analysis

Table 4 shows the discovery timeline:

The timeline divides into four phases. Physics establishes the framework between 1944 and 1971—widely celebrated, publicly available science.

Then a burst: between 1987 and 1994, at least four groups produce distinct criticality formulations. Peng et al. (1994) and Peters (1994) publish in the same year without cross-citation. This clustering coincides with increased computational capacity, which made DFA, Hurst analysis, and spectral methods tractable for large datasets.

Steady expansion follows from 2001 to 2013. ML, neuroscience, and engineering develop their own criticality methods. The 20-year gap between Kauffman (1993) and Saxe et al. (2013)—both analyzing neural computation—supports independent derivation rather than transfer.

Cross-domain awareness begins to emerge only in the 2010s. The 1990s clustering likely reflects

Table 4: Timeline of critical phenomena mathematics across disciplines

Year	Domain	Key Work
1935	Traffic Flow	Greenshields
1935	Linguistics	Zipf
1944	Statistical Physics	Onsager
1944	Seismology	Gutenberg & Richter
1940s	Conflict Studies	Richardson
1951	Hydrology	Hurst
1961	Climate	Stommel
1963	Finance	Mandelbrot
1966	Statistical Physics	Kadanoff
1971	Statistical Physics	Wilson (Nobel 1982)
1987	Complexity Science	Bak, Tang, Wiesenfeld
1993	Complexity Science	Kauffman
1994	Biomedical	Peng et al.
1994	Finance	Peters
1996	Climate	Schmidt & Mysak
2001	Machine Learning	Jaeger
2001	Materials Science	Sethna et al. (crackling noise)
2003	Neuroscience	Beggs & Plenz
2003	Linguistics	Ferrer i Cancho & Solé
2004	Cross-Domain Synthesis	Sornette
2004	Power Grids	Crucitti et al.
2004	Traffic Flow	Kerner
2005	Climate	Rahmstorf et al.
2007	Power Grids	Dobson et al.
2007	Urban Science	Bettencourt et al.
2008	Climate	Lenton et al.
2013	Machine Learning	Saxe et al.
2021	Climate/DL	Bury et al.

fields reaching analytical maturity and gaining computational tools simultaneously; we find limited evidence for coordinated dissemination.

The evidence is more consistent with natural convergence than with coordinated transfer. Each domain developed its own notation—no terminological copying is evident (Section). The 1990s clustering of DFA, Hurst analysis, and SOC applications raises the possibility of coordinated timing, but a more parsimonious explanation is simultaneous computational maturation: the processing power to run these analyses became available across fields in the same decade. The weight of evidence— independent notation, absent citations, institutional separation, domain-specific application—is most consistent with natural convergence, though the question remains open.

Correlation scaling emerges for critical transitions much as Fourier analysis emerges for periodic signals—the mathematics is dictated by the structure of the problem. Any complex system with interacting components can exhibit phase transitions, so researchers studying heart dynamics, market crashes, or traffic jams inevitably encounter correlation structure near critical points. The solution space is constrained: there are only so many ways to quantify whether correlations decay slowly or quickly. These domains all encountered bifurcation mathematics—saddle-node bifurcations producing critical slowing down, Hopf bifurcations generating oscillatory instabilities, pitchfork bifurcations breaking symmetry—whether or not practitioners recognized it as such.

Discussion and Implications

Independent discovery across multiple disciplines suggests criticality mathematics is a *widely applicable framework*—mathematical machinery that recurs wherever complex systems exhibit phase transitions. Insights transfer across domains: a power grid engineer can apply lessons from market crashes; an ML researcher can borrow intuition from physics. Multiple independent discoveries across decades, derivability from first principles, and peer-reviewed publication in at least five domains collectively suggest that the pattern of multiple independent derivations complicates any single claim of ownership over these techniques. Documenting the convergence pattern is itself a step toward the accessibility these techniques warrant.

Future Directions

Terminology standardization—developing shared vocabulary across domains—remains future work requiring sustained multi-domain collaboration. Cross-domain teaching materials presenting criticality from first principles would test whether unified instruction improves research outcomes. The convergence pattern invites further documentation: ecology (May’s 1977 ecosystem stability thresholds), social dynamics (opinion phase transitions), and epidemiology (transcritical bifurcations in disease dynamics) are candidates. As noted in the Limitations, a single round of AI-assisted search expanded our survey from nine to twelve domains, suggesting that systematic review would identify further instances.

Limitations

The functional correspondence we document is phenomenological—parameters detect similar critical signatures without necessarily being mathematically interconvertible in all cases. The rigorous equivalence $\alpha = H$ for fractional Gaussian noise (Section) does not automatically extend to all process types. Power-law behavior can arise from mechanisms other than criticality; our claim is that these techniques *converge* when applied to critical systems, not that all power laws indicate criticality. Sornette’s Dragon King theory (Sornette 2009; Sornette and Ouillon 2012) identifies important exceptions—predictable outliers generated by positive feedback rather than scale-free criticality.

Our citation analysis cannot rule out informal knowledge transfer. The quantitative evidence supports but does not prove independent derivation. We also note a scope limitation: broadening “criticality” beyond equilibrium phase transitions to include non-equilibrium SOC, long-memory stochastic processes, and algorithmic stability boundaries increases the risk that convergence becomes definitional rather than empirical. Our inclusion criterion is mathematical form—parameters that diagnose diverging correlation structure near a transition point—not the label “criticality.” We have attempted to mitigate definitional drift by classifying each domain’s relationship to the formal framework (Table 1), by distinguishing correlation-decay diagnostics from empirical scaling exponents (Table 3), and by restricting our strongest claims to cases where the mathematical correspondence is demonstrable.

Our survey expanded from an initial nine domains to twelve after a single round of AI-assisted literature search beyond the original manual survey. That one additional pass yielded three new domains (seismology, linguistics, urban scaling) suggests the convergence pattern is broader than documented here. A systematic review—beyond this paper’s scope—would likely identify further instances.

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Data Availability

Citation data were retrieved from the Semantic Scholar API (public tier). Analysis code and processed data are available from the corresponding author on request.

References

- Bak, P. and Tang, C. (1989). Earthquakes as a self-organized critical phenomenon. *Journal of Geophysical Research: Solid Earth*, 94(B11):15635–15637.
- Bak, P., Tang, C., and Wiesenfeld, K. (1987). Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4):381.
- Beggs, J. M. and Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35):11167–11177.
- Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., and West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences*, 104(17):7301–7306.
- Bury, T. M., Bauch, C. T., and Anand, M. (2021). Deep learning for early warning signals of tipping points. *Proceedings of the National Academy of Sciences*, 118(39):e2106140118.

541 Bury, T. M., Sujith, R. I., Pavithran, I., Scheffer, M., Lenton, T. M., Anand, M., and Bauch, C. T.
542 (2023). Predicting discrete-time bifurcation from noisy time-series data using deep learning.
543 *Nature Communications*, 14:5666.

544 Crucitti, P., Latora, V., and Marchiori, M. (2004). Model for cascading failures in complex networks.
545 *Physical Review E*, 69(4):045104.

546 Cvitanović, P. (1984). *Universality in Chaos*. Adam Hilger, Bristol.

547 Dakos, V., Carpenter, S. R., Brock, W. A., Ellison, A. M., Guttal, V., Ives, A. R., Kéfi, S.,
548 Livina, V., Seekell, D. A., van Nes, E. H., and Scheffer, M. (2012). Methods for detecting early
549 warnings of critical transitions in time series illustrated using simulated ecological data. *PLoS*
550 *ONE*, 7(7):e41010.

551 Dobson, I., Carreras, B. A., Lynch, V. E., and Newman, D. E. (2007). Complex systems analysis of
552 series of blackouts: Cascading failure, critical points, and self-organization. *Chaos*, 17(2):026103.

553 Ferrer i Cancho, R. and Solé, R. V. (2003). Least effort and the origins of scaling in human language.
554 *Proceedings of the National Academy of Sciences*, 100(3):788–791.

555 Fisher, M. E. (1974). The renormalization group in the theory of critical behavior. *Reviews of*
556 *Modern Physics*, 46(4):597.

557 Gardiner, C. W. (1983). *Handbook of Stochastic Methods for Physics, Chemistry and the Natural*
558 *Sciences*. Springer, 1st edition.

559 Greenshields, B. D. (1935). A study of traffic capacity. *Highway Research Board Proceedings*,
560 14:448–477.

561 Gutenberg, B. and Richter, C. F. (1944). Frequency of earthquakes in California. *Bulletin of the*
562 *Seismological Society of America*, 34(4):185–188.

563 Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society*
564 *of Civil Engineers*, 116:770–808.

- Jaeger, H. (2001). The "echo state" approach to analysing and training recurrent neural networks. Technical report, GMD Report 148, German National Research Center for Information Technology.
- Kadanoff, L. P. (1966). Scaling laws for ising models near t_c . *Physics Physique Fizika*, 2(6):263.
- Kantelhardt, J. W., Zschiegner, S. A., Koscielny-Bunde, E., Havlin, S., Bunde, A., and Stanley, H. E. (2002). Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A: Statistical Mechanics and its Applications*, 316(1-4):87–114.
- Kauffman, S. A. (1993). *The origins of order: Self-organization and selection in evolution*. Oxford University Press.
- Kerner, B. S. (2004). *The Physics of Traffic: Empirical Freeway Pattern Features, Engineering Applications, and Theory*. Springer.
- Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., and Schellnhuber, H. J. (2008). Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences*, 105(6):1786–1793.
- Mackey, M. C. and Glass, L. (1977). Oscillation and chaos in physiological control systems. *Science*, 197(4300):287–292.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *The Journal of Business*, 36(4):394–419.
- Mantegna, R. N. and Stanley, H. E. (1995). Scaling behaviour in the dynamics of an economic index. *Nature*, 376(6535):46–49.
- Merton, R. K. (1961). Singletons and multiples in scientific discovery: A chapter in the sociology of science. *Proceedings of the American Philosophical Society*, 105(5):470–486.
- Moore, C. and Mertens, S. (2011). *The Nature of Computation*. Oxford University Press.
- Onsager, L. (1944). Crystal statistics. i. a two-dimensional model with an order-disorder transition. *Physical Review*, 65(3-4):117.

- 590 Peng, C.-K., Buldyrev, S. V., Goldberger, A. L., Havlin, S., Sciortino, F., Simons, M., and Stanley,
591 H. E. (1994). Mosaic organization of dna nucleotides. *Physical Review E*, 49(2):1685.
- 592 Peng, C.-K., Havlin, S., Stanley, H. E., and Goldberger, A. L. (1995). Quantification of scaling
593 exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*, 5(1):82–87.
- 594 Peters, E. E. (1994). *Fractal market analysis: applying chaos theory to investment and economics*.
595 John Wiley & Sons.
- 596 Rahmstorf, S., Crucifix, M., Ganopolski, A., Goosse, H., Kamenkovich, I., Knutti, R., Lohmann,
597 G., Marsh, R., Mysak, L. A., Wang, Z., and Weaver, A. J. (2005). Thermohaline circulation
598 hysteresis: A model intercomparison. *Geophysical Research Letters*, 32(23):L23605.
- 599 Richardson, L. F. (1960). *Statistics of Deadly Quarrels*. Boxwood Press, Pittsburgh.
- 600 Saxe, A. M., McClelland, J. L., and Ganguli, S. (2013). Exact solutions to the nonlinear dynamics of
601 learning in deep linear neural networks. In *International Conference on Learning Representations*
602 (*ICLR*). Presented at ICLR 2014.
- 603 Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H.,
604 Van Nes, E. H., Rietkerk, M., and Sugihara, G. (2009). Early-warning signals for critical transi-
605 tions. *Nature*, 461(7260):53–59.
- 606 Schmidt, G. A. and Mysak, L. A. (1996). The stability of a zonally averaged thermohaline circula-
607 tion model. *Tellus A*, 48(1):158–178.
- 608 Sethna, J. P., Dahmen, K. A., and Myers, C. R. (2001). Crackling noise. *Nature*, 410(6825):242–250.
- 609 Sornette, D. (2004). *Critical Phenomena in Natural Sciences: Chaos, Fractals, Selforganization*
610 *and Disorder: Concepts and Tools*. Springer, 2nd edition.
- 611 Sornette, D. (2009). Dragon-kings, black swans and the prediction of crises. *International Journal*
612 *of Terraspace Science and Engineering*, 2(1):1–18.
- 613 Sornette, D. and Ouillon, G. (2012). Dragon-kings: Mechanisms, statistical methods and empirical
614 evidence. *The European Physical Journal Special Topics*, 205(1):1–26.

- 615 Stommel, H. (1961). Thermohaline convection with two stable regimes of flow. *Tellus*, 13(2):224–
616 230.
- 617 Wang, Z. and Mysak, L. A. (2006). Glacial abrupt climate changes and Dansgaard-Oeschger
618 oscillations in a coupled climate model. *Paleoceanography*, 21(1):PA1009.
- 619 Wilson, K. G. (1971). Renormalization group and critical phenomena. *Physical Review B*, 4(9):3174.
- 620 Wissel, C. (1984). A universal law of the characteristic return time near thresholds. *Oecologia*,
621 65(1):101–107.
- 622 Zipf, G. K. (1949). *Human Behavior and the Principle of Least Effort*. Addison-Wesley, Cambridge,
623 MA.